

The re-emergence of natural products for drug discovery in the genomics era

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Abstract | Natural products have been a rich source of compounds for drug discovery. However, their use has diminished in the past two decades, in part because of technical barriers to screening natural products in high-throughput assays against molecular targets. Here, we review strategies for natural product screening that harness the recent technical advances that have reduced these barriers. We also assess the use of genomic and metabolomic approaches to augment traditional methods of studying natural products, and highlight recent examples of natural products in antimicrobial drug discovery and as inhibitors of protein–protein interactions. The growing appreciation of functional assays and phenotypic screens may further contribute to a revival of interest in natural products for drug discovery.

Chemical space

The multidimensional space occupied by all chemical compounds.

Pharmacophores

Molecules with atoms or chemical groups in a spatial arrangement known or predicted to be responsible for specific biological activity.

Stereochemistry

The spatial arrangement of atoms in a molecule.

There is an increasingly powerful case for revisiting natural products for drug discovery. Historically, natural products from plants and animals were the source of virtually all medicinal preparations and, more recently, natural products have continued to enter clinical trials or to provide leads for compounds that have entered clinical trials, particularly as anticancer and antimicrobial agents^{1–3}. A detailed analysis of new medicines approved by the US Food and Drug Administration (FDA) between 1981 and 2010 (REF. 4) revealed that 34% of those medicines that were based on small molecules were natural products or direct derivatives of natural products — including the statins, tubulin-binding anticancer drugs and immunosuppressants^{5–8}.

This contribution of natural products seems impressive, especially against a backdrop of the diminished focus on natural products by the major pharmaceutical companies. There have been several reasons for the lack of enthusiasm for using natural products for drug discovery in industry. There are legitimate concerns arising from the United Nations Convention on Biological Diversity that seeks to regulate international access to natural products, but these can be dealt with by following accepted best practices (BOX 1). Of greater concern was the belief that natural products are somehow incompatible with drug discovery approaches that are based on high-throughput screening (HTS) directed at molecular targets⁹, and concerns about the difficulties associated with the repeated isolation of known compounds and with synthesizing natural products during pharmaceutical manufacture¹⁰. This was coupled with the as yet unfulfilled

expectation that combinatorial chemistry techniques could provide all the chemicals needed for successful lead discovery^{4,11,12}. Many large screening collections have been disappointing in practice, and it is recognized that diversity within biologically relevant ‘chemical space’ is more important than library size. A retrospective analysis of one company’s HTS campaigns indicated that the selection of plates containing natural products would have significantly improved hit rates¹¹. To a certain extent, compound libraries are being designed to mimic more closely the chemical properties of natural products^{10,13–17}, although natural products themselves are generally overlooked.

Natural product collections exhibit a wide range of pharmacophores and a high degree of stereochemistry, and these properties are expected to contribute to the ability of such collections to provide hits — even against the more difficult screening targets, such as protein–protein interactions¹⁰. However, natural products may have the additional advantage over synthetic compounds of being natural metabolites: compounds that are successful as drugs have been suggested to have the property of ‘metabolite-likeness’ (REF. 18). This means that such compounds are not only biologically active but also likely to be substrates for one or more of the many transporter systems that can deliver the compounds to their intracellular site of action.

A high degree of bioavailability could be particularly important if the trend towards more functional assays continues. With the development of high-content screening and other phenotypic assays that use cells or

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Box 1 | The United Nations Convention on Biological Diversity and the Nagoya Protocol

The United Nations [Convention on Biological Diversity](#) (CBD) arose from the Rio Earth Summit in Brazil in 1992. It sets out the expectations around access to, and the use of, biodiversity ('genetic resources') across national boundaries. Broadly, the CBD states that:

- Countries have sovereign rights over the genetic resources in their territories
- Access to genetic resources by foreign groups requires prior informed consent from the appropriate authority in the source country
- Access on mutually agreed terms should be facilitated by the source country
- Benefits from the use of genetic resources should be shared in a fair and equitable way with the source country
- The source country should be involved in relevant research on the genetic resources, where possible, and benefit from technology transfer

The CBD was signed by most countries in the world (194 to date), and has been extensively ratified (with the notable exception of the United States). However, because the CBD requires individual countries to adopt suitable laws and regulations to implement its principles, the impact of the CBD has been varied.

The CBD did not have specific recommendations that addressed the use of traditional knowledge from one group of people by other groups or companies. This was the subject of the [Nagoya Protocol](#), which was adopted in October 2010 by the Conference of the Parties to the CBD. This Protocol gives detailed suggestions that cover access and benefit-sharing with respect to natural products and traditional knowledge. The Protocol has been signed by 92 countries (as of November 2014), but is only ratified by 30 countries, with Norway being the first 'developed' country to do so. Fifty countries have to adopt the Protocol before it has legal force. However, it can be regarded as a practical guide for those working on biodiversity and making use of traditional knowledge.

For further discussions on the impact of the CBD and the workings of the Nagoya Protocol, see REFS 2,272,273,274.

model organisms^{19,20}, test-compound bioavailability becomes important earlier in the screening cascade than if the primary assay is a molecular one, such as an isolated protein assay. There is also increasing interest in the clues from traditional uses of plant extracts — for example, those in traditional Chinese medicine (TCM) — to guide new drug discovery. There are several examples of leads that are based on TCM^{21–23}, although it is still unclear whether or not these are useful signposts²⁴.

Given the recent technological advances, and the apparently growing appreciation of functional assays and phenotypic screens, will natural products come back into favour among drug discovery scientists? In this Review, we describe recent efforts to optimize the use of natural products and their derivatives in drug discovery. We first discuss advances in the development of screening libraries based on natural product extracts, fractions prepared from extracts and purified compounds, and the use of chemical scaffolds based on natural products. We then consider the use of metabolomics and metagenomics in identifying new classes of natural products, such as those from previously non-cultivated bacterial strains. Finally, we review the progress in the discovery of potential drugs derived from natural products, in particular citing examples of products with novel antimicrobial properties and products that target protein–protein interactions.

Advances in natural-product-based screening

The traditional approach of bioassay-guided isolation of natural products is being modified to take advantage of technological advances, to accommodate current understanding in medicinal chemistry, and to explore biologically relevant chemical space via cheminformatic approaches to the design of libraries.

Improving the applicability of natural-product-based screening collections. Traditionally in natural-product research, concentrated extract samples were screened in bioassays²⁵. Such extracts are complicated mixtures. All of the components of the extract may reach the biological target in the assay, but some components may be in concentrations that are too low to have measurable effects, or the signal from the assay may be confounded by interference or nuisance compounds, or by the additive or synergistic effects of several compounds. Isolating each compound from a crude extract in advance of screening, however, is likely to be too onerous and uneconomical to be feasible for large numbers of samples.

A first step towards simplifying extracts and making them more suitable for use in bioassays is to remove compounds that are likely to cause artefacts: polyphenolic tannins are the usual suspects in plant extracts²⁶. Then, fractions of reduced complexity can be prepared for screening, allowing the scale to be miniaturized and the speed enhanced. One necessity arising from pre-fractionation is that of a repository, from which small quantities of an increased number of fractions can be biologically tested using HTS. The use of simplified fractions, together with sensitive NMR techniques (BOX 2), has addressed the isolation and structure-elucidation bottleneck. Additionally, as fractions are prepared by a chromatographic method, subsequent chromatography on existing fractions is more likely to be achievable, avoiding the previous danger of not finding the responsible constituent in active crude extracts.

Numerous pre-fractionation strategies have been reported^{27–36}, and these are summarized in TABLE 1. Pre-fractionation HTS strategies have been shown to give higher hit rates. From a microbial natural-product library comprising 1,882 active cultures, 79.9% of the activities that were found by nine HTS campaigns (against three

Metabolomics

The systematic, qualitative and quantitative analysis of all metabolites contained or produced in an organism at a specific time and under specific conditions.

Metagenomics

The sequencing and analysis of DNA from environmental samples without the need for culturing individual clonal organisms.

Box 2 | **Advances in NMR techniques for structural elucidation**

Driven by the enhanced sensitivity of NMR microcryoprobes, the ability to elucidate chemical structures of natural products using NMR spectroscopy has now reached the stage where structures can be solved using very small quantities (in the microgram range). At such sensitivity, synthesis of the isolated compounds will be required to allow subsequent full biological evaluation. Examples of compounds and the mass from which their structure was solved are as follows:

- 9-O-desmethylkabiramide B (620 μg)^{275,276}
- Sanguinamides A and B (390 μg and 190 μg , respectively)^{275,276}
- Phorbaside F (7.5 μg)^{275,276}
- Hemi-phorboxazole A (16.5 μg)^{275,276}
- Muironolide A (90 μg)^{275–277}
- Combretasides A–G (new acetylated dammarane-type bisdesmosides) (29–187 μg)²⁷⁸
- Beilschmiedic acids O (41.9–220 μg)²⁷⁹
- One previously known (120 μg), and three new (60 μg , 180 μg and 5 μg) stilbenoids^{279–281}

Structural elucidation of novel natural products demands two-dimensional NMR experiments. Hyphenated (combined) techniques such as high-performance liquid chromatography (HPLC)–NMR use stopped-flow, and thus allow for smaller quantities to be used²⁸².

One structural difficulty associated with natural products is that they are frequently isolated as a single diastereoisomer²⁸³, meaning only one set of NMR measurements is possible. Although computational (*ab initio*) calculation of NMR chemical shifts is a powerful tool for assigning stereochemistry to molecules^{284,285}, having just one data set for a single diastereoisomer provides an additional challenge in determining the stereochemistry²⁸³. A probability method of determining stereochemistry that is based on conformational analysis of diastereoisomers has been successfully demonstrated for 36 structures²⁸³. Complex mixtures of metabolites can be elucidated using another two-dimensional technique: *J*-resolved NMR. This method can be used to resolve overlapping signals by separating the effect of *J*-coupling⁷² from the effect of chemical shifts. The magnitude of the *J*-coupling signal provides information on bond distance and bond angles — information that can be correlated to a compound's stereochemical features.

ion channels, three kinases and three unspecified screens) were observed only in the fractions, whereas only 12.5% of the activity was found in the crude extract. Only 7.6% of the activity was observed in both the crude extract and the fractions²⁹. A very similar result was achieved using the simpler 'four fractions per extract' approach. Of the 1,700 active fractions from 11 HTS screens (including whole-cell and/or enzyme- or protein-based assays), 80% of the primary-screen hits from pre-fractionated samples were not observed in assays of the associated crude extract²⁸.

Characteristics of screening libraries based on natural products. Biological structural space has been defined as the protein-binding sites for potential ligands³⁷. If screening is viewed as the search for molecules in chemical space that are 'complementary' to the biological space for macromolecular targets involved in disease pathology, a key issue is the biological relevance of the structural diversity in chemical libraries³⁷.

The majority of screening libraries that are based on synthetic compounds are composed of compounds that have drug-like or lead-like properties with regard to absorption, distribution, metabolism, excretion and toxicity (ADMET) characteristics³⁸, with compounds that have reactive groups or other problematic chemical substructures removed³⁹. This results in a subset of chemical diversity that is aimed towards cell permeability

and oral bioavailability, but which still does not address the issue of biologically relevant chemical space. As a result, the screening of large numbers of compounds is typically still necessary to identify active compounds. The reverse approach is to start with a set of biologically relevant chemicals and then to apply drug-like or lead-like ADMET filters. Natural products have a distinct advantage in this approach, as they inherently fall in regions of biologically relevant chemical space^{13,40–42} (FIG. 1).

The modern drug-discovery approach of pre-filtering screening libraries for drug- or lead-likeness can be applied to natural-product fraction libraries. In this regard, the Wyeth fractionation procedure recognized the value in collecting fractions that were most likely to contain compounds with drug-like properties, and so the highly polar compounds, the early-fractionation material consisting of media components, and the late-eluting lipophilic portion were discarded²⁹. Natural product samples can be enriched for drug-like properties by concentrating on fractions that contain compounds in the appropriate range of lipophilicity³³. The closer a fraction library can approach the typical industry definition of a lead molecule with regard to physicochemical properties, the greater the likelihood of identifying a starting point that is attractive from a medicinal chemistry perspective. In order to align natural-product screening with the same physicochemical profile of synthetic screening libraries, a solid-phase extraction (SPE) method may be used to remove high logP components and provide logP < 5 constituents with the required physicochemical profile before fractionation^{33,42}.

Natural products interrogate a different, wider and more drug-like chemical space than do synthetic derivatives^{43,44}. Furthermore, it has been shown that 83% of core ring scaffolds that are present in natural products were absent from commercially available molecules and screening libraries¹⁸. It was concluded that including molecules with a natural-product-like scaffold into a screening library would increase hit rates¹⁸.

Besides natural-product-likeness based on chemical structure, similarity measures can also be based on physicochemical properties¹⁶. The chemical space navigation tool ChemGPS-NP was used to analyse natural products and bioactive medicinal chemistry compounds in the **WOMBAT** database¹⁶. There was limited overlap between the biologically relevant chemical space covered by natural products and the biologically relevant chemical space covered by medicinal chemistry compounds, showing that natural products cover parts of the chemical space that lack representation by medicinal chemistry compounds. Such natural products may be useful novel leads¹⁶. A comparison of the Euclidean distances (EDs) in chemical space between approved drugs included in the **GVK BIO** drug database and natural products from the **Dictionary of Natural Products** identified many drug–natural-product pairs. This analysis revealed that 99.5% of all drugs have a natural-product 'neighbour' that is closer than ED = 10, and that 85% of drugs have a natural-product neighbour closer than ED = 1, where ED = 0 is an exact match. This analysis indicated that natural products with very short EDs to any approved drug may be a potential lead against the same target

Drug-like

Sharing certain characteristics — such as size, shape and solubility in water and organic solvents — with other molecules that act as drugs.

LogP

Logarithm of the octanol–water partition coefficient, which is a measure of a drug's lipophilicity. Defined as the ratio of un-ionized drug distributed between the octanol and water phases at equilibrium. Higher values imply greater lipophilicity.

Table 1 | **Pre-fractionation strategies**

Institute	Number of fractions	Number of samples	Method of generating fractions	Refs
MerLion Pharmaceuticals	4 HPLC fractions per sample	≤120,000	C18 HPLC	28
Sequoia Sciences	40 HPLC fractions on 5 subfractions (200 fractions per sample)	36,000 fractions, with each well containing approximately 1–5 compounds	Organic extract: silica FC (to give 4 fractions), followed by HPLC (to give 40 fractions); aqueous extract: pre-treated C18 FC, polyamide chromatography, molecular weight filter, then HPLC (to give 40 fractions)	27
Wyeth	10 HPLC fractions per sample	≤6,500	C18 HPLC	29
Ireland Lab (University of Utah, USA)	20 HPLC fractions from 4 subfractions (80 fractions per sample)	15,360	Synthetic adsorbent separation, followed by C18 HPLC	30
Guy and Yan Group (St Jude Children's Research Hospital, Tennessee, USA)	24 fractions	≤62,000	Polyamide FC, followed by C18 HPLC	31
RIKEN (Japan)	Up to 325 fractions per sample	~6,500 semi-purified natural-product fractions	LC–MS	32
Quinn Lab (Eskitis Institute, Griffith University, Brisbane, Australia)	11 HPLC fractions per sample	202,983	Oasis HLB, followed by C18 HPLC	33
Watnick Lab (Children's Hospital, Boston, Massachusetts, USA)		39,000	Pre-fractionated	34
Biomedical Information Research Center (Japan)		123,599	Fractionated to give crude metabolites	35
Linington Lab (Santa Cruz, California, USA)	6 fractions per sample	3,120	Solid-phase extraction C18 reverse-phase chromatography	36

FC, flash chromatography; HLB, hydrophilic–lipophilic balance separation; HPLC, high-performance liquid chromatography; LC–MS, liquid-chromatography–mass-spectrometry.

as that of the drug¹⁶. These property-based similarity calculations can identify structurally dissimilar compounds that have neighbours with similar properties¹⁶.

Another strategy towards creating a library based on natural products relies on a structure-based approach to identify biologically relevant compound classes and scaffolds¹³. Biology-oriented synthesis follows the principle that, over the course of evolution, nature conserves the scaffolds of natural products and of protein backbones. In biology-oriented synthesis, compound classes and scaffolds are therefore selected primarily on the basis of biological compatibility. The particular compound class is decided by either structural classification of natural products (SCONP) or protein structure similarity clustering (PSSC), or both are used in combination. This approach has led to 'natural-product-derived' libraries (where the scaffold is identical to the natural-product scaffold) and 'natural-product-inspired' libraries (where the scaffold is closely related to natural-product scaffolds). Natural-product-inspired compound collections that have been synthesized according to the logic of biology-oriented synthesis have proven to be enriched in bioactivity over typical collections of synthetic compounds: hit rates of

0.2–1.5% have been found against molecular targets such as phosphatases and in cell-based assays¹³. For example, combined PSSC and SCONP were used to generate a library of 483 decalins to discover selective inhibitors of the corticosteroid 11 β -dehydrogenase isozyme 1 (11 β HSD1). PSSC identified similarities between the M-phase inducer phosphatase 1 CDC25A, acetylcholinesterase and 11 β HSD1. Subsequently, the CDC25A inhibitor dysidiolide and the 11 β HSD1 ligand glycyrrhetic acid were analysed using SCONP to identify a 1,2-dehydrodecalin scaffold, and the resultant synthetic library based on this scaffold contained three highly isoenzyme-selective 11 β HSD1 inhibitors with nanomolar potency¹³.

The use of natural products as a guiding structure, and the chemical transformation of scaffolds derived from (or embedded within) the guiding structure to create natural-product-like libraries, has been analysed with respect to variation of the two main lead-like descriptors: molecular mass (MM) and lipophilicity (logP)⁴⁵. This analysis revealed that, although there are many strategies to create natural-product-like libraries, the impact of chemical transformations on physicochemical properties in the generation of libraries has not been uniformly

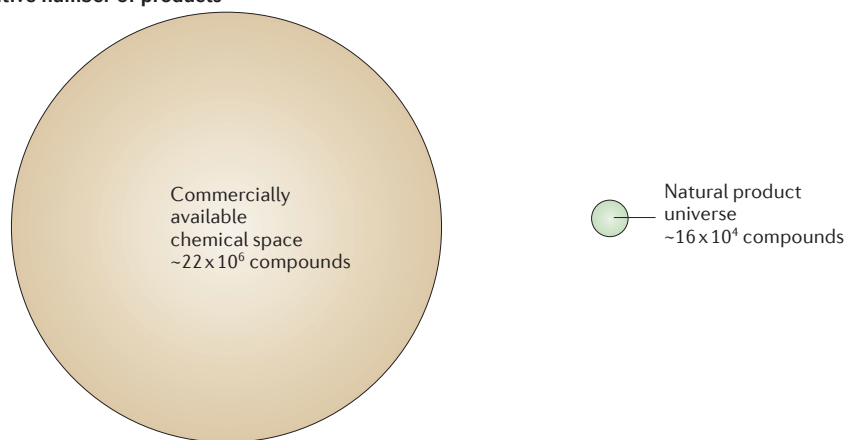
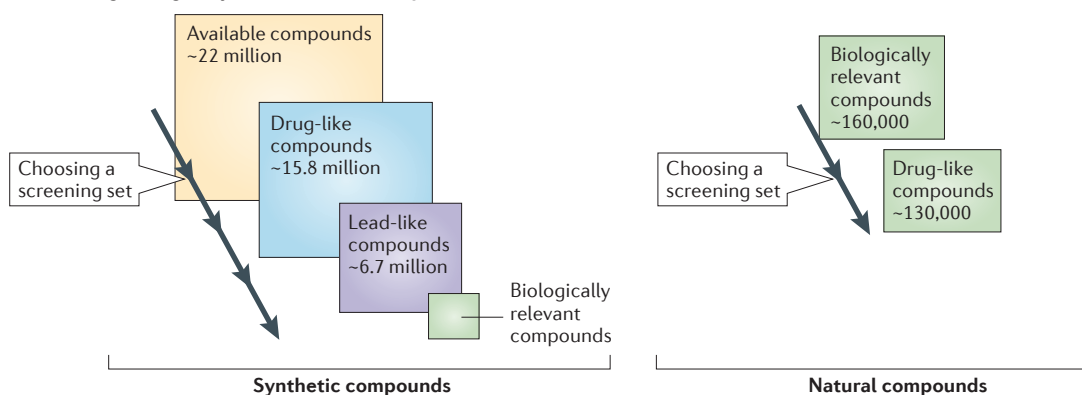
a Relative number of products**b Choosing biologically relevant chemical space**

Figure 1 | Biologically relevant chemical space is better covered by natural products than by synthetic compounds.

a | There are 22,724,825 commercially available compounds in the ZINC database²⁸⁶, as represented by the large brown circle. By comparison, there are around 160,000 unique natural products in the *Dictionary of Natural Products*, represented in approximate scale by the small green circle. **b** | Compounds that are biologically relevant by definition 'hit' a biological target. The ZINC database (represented by the yellow square) is designed to allow virtual screening of listed compounds, which are in ready-to-dock three-dimensional formats. The typical size of physical screening libraries ranges from ~100,000 compounds to millions of compounds. Screening libraries may be chosen to conform to Lipinski's rule of five (represented by the blue square) or to be lead-like (represented by the purple square) with reduced molecular mass and lipophilicity. The ZINC database's drug-like subset consists of 15,798,630 compounds, whereas the same database's lead-like subset comprises 6,687,370 compounds. Considerable effort, knowledge and expertise is required to select a specific screening subset that is enriched with biologically relevant compounds (represented by the small green square). By contrast, all natural products occupy biologically relevant chemical space (as described by protein fold topology)^{40,41}, and around 80% of natural products conform to the rule of five and are drug-like⁴².

Rule of five

A set of criteria, known collectively as Lipinski's 'rule of five', that identify several key properties that should be considered for compounds with oral delivery in mind. These properties are as follows: molecular mass < 500 Da, cLogP < 5, number of hydrogen-bond donors < 5 and number of hydrogen-bond acceptors < 10.

addressed. During the lead-optimization process, an increase in MM and logP is generally observed^{46,47}, and thus the optimum property cut-off values when identifying natural-product lead-like libraries should respect the following conditions: MM < 350 Da; and $-1 < \log P < 3$. If library generation is to fit within the modern drug discovery paradigm, the library members should all be within biologically relevant chemical space and allow for increases in MM and logP during lead optimization: that is, the library members must be within the purple square of FIG. 1. If the guiding natural product falls in the lead-like region, the guiding natural product can be used as a starting point for the preparation of a library with skeletal and stereochemical variations, provided the generated molecules remain in the lead-like region⁴⁵. However, if the

guiding natural product has a MM > 350 Da, it is necessary to reduce the structural complexity^{45,48}, and if the guiding natural product has a MM < 350 and logP > 3, it is necessary to reduce lipophilicity.

The structures of many natural products are available in searchable databases (TABLE 2) and some of these have been used in virtual-screening campaigns. These approaches can help optimize the use of compounds that are available in only very small amounts⁴⁹.

Applying 'omics' to natural products

Metabolomics arose from the concepts of metabolic profiling and has the goal of qualitatively and quantitatively analysing all the metabolites that are contained in an organism at a specific time and under specific conditions.

Table 2 | Natural-product databases that can be used for virtual screening campaigns

Database	Number of entries	Additional information	Refs
Super Natural II	355,000	2D structures; vendor information for over 215,000 compounds	*
Universal Natural Product Database	197,201	3D structures assembled from other available Chinese databases	289
Chinese Natural Product Database	53,000	Has been used in a virtual screen for PPAR- γ agonists	290
Drug Discovery Portal	40,000	Not all natural products, but all based on available samples	49
iSMART	20,000	Based on components from traditional Chinese medicines	291, 292
Database from historical medicinal plants, DIOS	6,702	Successfully used in several virtual screening campaigns	293
AfroDb	1,000	Compounds from African medicinal plants	294
NuBBE	640	Compounds from Brazilian sources	295 [‡]

2D, two-dimensional; 3D, three-dimensional; iSMART, integrated systems biology-associated research with traditional Chinese medicine; PPAR- γ , peroxisome proliferator-activated receptor- γ . *See the Super Natural II database. [‡]See the NuBBE database.

This approach ultimately allows indirect monitoring of gene function and the biochemical status of an organism. A combination of metabolomics and genomics can be used to optimize a biosynthetic pathway to selectively produce biologically active secondary metabolites^{50–52}.

Profiling and isolation using metabolomics. The application of metabolomics in natural-product research began almost a decade after it was already well established in the fields of biomedical⁵³ and agricultural research⁵⁴ with the introduction of photo-diode arrays⁵⁵, along with HRFTMS⁵⁶ (high-resolution Fourier-transform mass spectrometry) detectors that were coupled with HPLC (high-performance liquid chromatography). By the beginning of the millennium, high-throughput sequencing was emerging and there was a shift from pure genetic research to the elucidation of gene function and expression^{57,58}. The ability of an organism to produce secondary metabolites is a phenotype and these metabolic phenotypes have been researched using metabolomics (which, in such cases, may also be known as phenomics)^{59,60}.

The basic analytical techniques HRFTMS^{61,62} and NMR spectroscopy^{63–66} are being used to dereplicate and quantify known metabolites against novel natural products^{67,68} (BOX 2). Dereplication is a massive task^{69,70} because secondary metabolites often have complex and diverse atomic arrangements, which makes elucidating their structure challenging. A combination of analytical methods that includes ultraviolet spectroscopy, tandem mass spectrometry and NMR spectral data is used to ensure the correctness of the identification of the basic structure of dereplicated compounds in crude samples. Ultraviolet data can only be used for the analysis and dereplication of chromophore-containing metabolites, whereas in mass spectrometry there is the risk of missing poorly ionizing compounds that are only detectable in one ionization mode. For example, phenolic and anthraquinone compounds poorly ionize in the positive ion mode but ionize very well in the negative ion mode⁶⁸, whereas the opposite can be said for alkaloids.

Secondary metabolites can also be found in a wide range of concentrations. Reliable, robust, selective and high-resolution analytical methods are therefore required for identifying and quantifying the multiple chemical groups of natural products. Standard NMR methods or pulse sequences are being reconceived to identify a complex mixture of metabolites in a typical natural-product extract. One method of doing this is two-dimensional J-resolved NMR^{71,72} (BOX 2).

Along with metabolomics-guided fractionation tools, it is possible to pinpoint active components at the first fractionation step, as well as predict which structures might be bioactive^{68,73}. The use of metabolomics could help prioritize fractions for further purification, which should save time and resources in isolating the target compounds (FIG. 2). There is rapid progress on pioneering information technologies^{67,74–77} for quick and efficient analysis of large data sets of natural-product libraries⁷⁸. These harness innovative approaches to analyse mass spectrometry data — for example, MZmatch^{79–81} and XCMS^{82,83} — that can be used with differential expression analysis and with online databases (such as [ChemSpider](#), [MarinLit](#) and the Dictionary of Natural Products) or in-house databases⁵⁰. Differential expression analysis⁷⁹ involves a series of processes, including nonlinear retention-time alignment of compounds that are unique to the sample, followed by matched filtration of authentic peaks in mass spectra to the sample and against noise and background peaks (such as those resulting from solvents or media). Finally, peaks are detected and known metabolites are matched to compounds using a database such as the Dictionary of Natural Products. Unique features of compounds can be deduced to forecast biological activity through pattern recognition. Multivariate analysis using statistical tools — such as unsupervised principal components analysis, supervised discriminant function analysis and Z-score analysis — can be used for pattern recognition programmes, which may predict the biological activity that results from particular features of compounds^{67,73,77,84,85}.

Dereplication

The process of using spectroscopic methods to identify known metabolites during the preliminary screening stage and eliminating further isolation work on already well-studied natural products.

Metabolomics can also be applied to optimize fermentation, and to detect and maintain the production of interesting secondary metabolites during cultivation and production processes, hence assisting in enhancing the biosynthesis of desired compounds. Along with the OSMaC (one strain, many compounds) approach^{86,87}, metabolomics can be used to explore and statistically validate relationships between culture methods, diversity, bioactivity and metabolome evolution in a microbial isolate^{56,68}. During scale-up, production processes in a small-scale fermenter can be optimized using real-time metabolomics^{88,89}. A detailed real-time metabolome-pathway analysis can fully characterize the intermediates, by-products and degradants within a metabolic flux system⁹⁰. Thus, metabolic bioproduction systems can be engineered; for instance, the stability of the production of the desired components when changing certain fermentation parameters can be checked and maintained⁹⁰.

Traditional herbal medicine such as TCM entails the use of a mixture of plant products, and metabolomics has facilitated the study of the effects of these complex mixtures on complex biological systems^{21,91–93}. Metabolic-profiling strategies also have practical uses in research and development and in the quality control of raw- and end-product TCM products⁹⁴. Tea-sourced⁹⁵ health products and Chinese patent medicine samples have been analysed using chemometric-guided HPLC^{95–97} to provide quantitative information on specific individual compounds from within overlapping profiles of co-eluting compounds. This metabolomics approach of metabolic fingerprinting, coupled with chemometric analysis, has been described in the analysis of TCM products and of raw materials from *Angelica sinensis*^{98,99}. Metabolic profiling with electrospray ionization quadrupole–time-of-flight mass spectrometry (ESI–TOFMS), another metabolomics-based analytical method, was also used for the efficient isolation of a new antitumour saponin from *Panax ginseng*, a popular TCM¹⁰⁰.

Metabolomics data can be mined to postulate biosynthetic precursors that can be useful in engineering pathways to increase the yield of the functional natural product. For example, in the production of ephedrine bronchodilators, targeted metabolic profiling and comparative biochemical analyses revealed benzaldehyde to be an important precursor of phenylpropylamino alkaloids produced in *Ephedra* spp.¹⁰¹. Moreover, through metabolic profiling it was possible to investigate the biosynthesis of tanshinone and to increase the expression of one of the tanshinone-synthesizing enzymes, SmCPS (*Salvia miltiorrhiza* copalyl diphosphate synthase), in *S. miltiorrhiza* hairy root cultures, thereby increasing overall tanshinone production¹⁰². Tanshinones have shown promise as therapeutic agents for oxidative stress injury in neurodegenerative, cardiovascular¹⁰³ and cerebrovascular disorders¹⁰⁴. For instance, tanshinones have been found to delay the development of ischaemia induced by myocardial infarction in rats by decreasing infarct size and improving systolic function¹⁰³. In zebrafish and rodent models of seizures, tanshinones — in

particular, one of the active tanshinones, tanshinone IIA — were shown to suppress seizure activity¹⁰⁴. Tanshinone IIA is already prescribed in China to treat cerebral ischaemia.

Metagenomics or gene manipulation for synthetic pathways. Genomics and metagenomics have become part of the workflow (FIG. 2) in the target-directed search not only for new bioactive secondary metabolites^{51,105–116} but also for novel microorganisms^{51,105,107,114,117,118} from under- or un-explored geographical areas^{119–122}. Heterologous expression of non-ribosomal and cryptic gene clusters (also known as silent gene clusters) is also currently being used as a strategy to explore the production of the bioactive secondary metabolites^{113,123–130}. Strains of microorganisms that are currently not cultivable pose a challenge as they cannot be used to produce the target compounds at the yields that are required for further bioassays and elucidation work. However, successful expression of biosynthetic enzymes and desired secondary metabolites has been achieved by heterologous expression in cultivable host strains, such as *Escherichia coli*^{128,131–141}, *Saccharomyces cerevisiae*^{142–147} and *Streptomyces coelicolor*^{126,130,148}.

Metagenomics has led to the exploration of the heterologous expression of biosynthetic gene clusters in microorganisms. Indeed, metagenomic approaches have now become essential tools for elucidating, defining and controlling biosynthetic pathways¹³¹. A key goal of these approaches is to develop an efficient ‘plug and play’ heterologous gene expression platform that — through ‘genome mining’ — will facilitate the production of particular target secondary metabolites^{149,150} or a plethora of new natural products^{111,151–154}.

Bacterial or fungal polyketides^{125,129,153,155–157} are chemically diverse and some have displayed antibiotic activity by inducing ribosomal frameshift errors¹⁵⁸ — a strategy thought to avoid antibiotic-resistance mechanisms¹⁵⁹. However, owing to the intrinsic molecular complexity of polyketides, chemical synthesis may not be commercially viable. Gene clusters are involved in every step of the production of biologically active polyketides¹⁶⁰, and manipulating biosynthetic gene clusters can allow more economical and efficient production of chirally pure polyketide compounds¹⁶¹. Two examples, epothilone¹⁶² and lomaivitin¹⁶³, are presented in FIG. 3.

Epothilones from the myxobacterium *Sorangium cellulosum* are potential anticancer drugs that act as microtubule disruptors, similarly to taxanes¹⁶⁴. Several epothilone analogues are currently undergoing clinical trials: patupilone (also known as EPO-906 or epothilone B) has been through Phase III trials in the United States, and Phase III trials are ongoing in the United Kingdom, Spain and Greece for ovarian cancer therapy. Ixabepilone (Ixempra; Bristol-Myers Squibb), an analogue of epothilone B, has been approved for the treatment of breast cancer¹⁶⁵. The biosynthetic gene clusters of epothilones have been widely studied^{166,167}. Recently, a 56-kb epothilone biosynthetic gene cluster was reassembled using unique restriction sites (which allowed for future module interchangeability) in the guanine- and cytosine-rich host *Myxococcus xanthus*¹⁶⁸.

One strain, many compounds

(OSMaC) An approach to activate metabolic pathways — by altering cultivation parameters, through co-cultivation or through the use of enzyme inhibitors or elicitors — that can be combined with genomics scanning.

Cryptic gene clusters

Genes that are normally not expressed (are ‘silent’) and do not translate into a phenotypic difference but that can become visible in a different environment to generate phenotypic diversity.

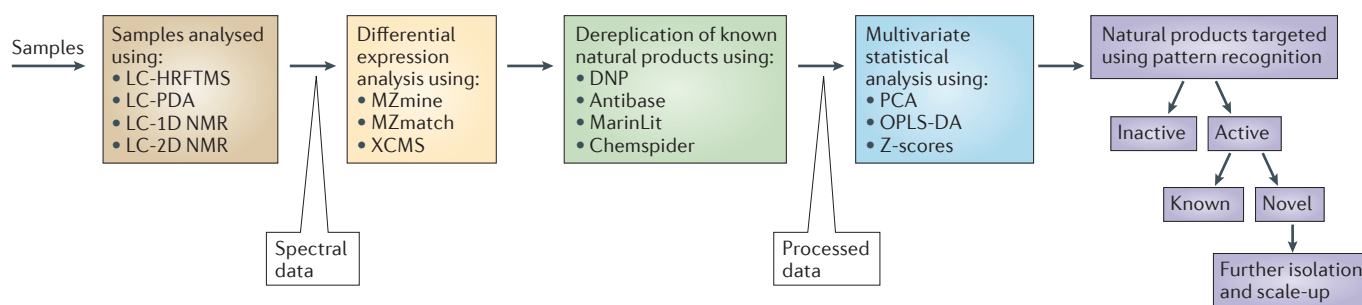


Figure 2 | Metabolomics data workflow in natural product research. Samples are submitted to LC-HRFTMS (liquid chromatography–high-resolution Fourier-transform mass spectrometry), LC-PDA (liquid chromatography–photodiode array) and LC-1D/2D NMR (liquid chromatography–one dimensional/two dimensional NMR spectroscopy) analysis. The mass spectrometry data are further processed using differential expression analysis software such as MZmine, MZmatch and XCMS. This software is coupled to databases such as the [Dictionary of Natural Products](#) (DNP), [AntiBase](#), or [MarinLit](#) to dereplicate known natural products against the novel secondary metabolites. Pre-collected LC-PDA and LC-1D/2D NMR data confirm the dereplication results. The processed data are subjected to multivariate analysis using both PCA (principal component analysis) and/or OPLS-DA (orthogonal partial least squares discriminant analysis). The results are then plotted in S-plots and heat maps. Through pattern recognition, inactive versus active and known versus novel natural products are sorted to define the natural products that will be targeted for further isolation and scale-up work.

Genome mining has been used to characterize orphan gene clusters in *Salinispora tropica* CNB-440 (REFS 150, 169). Moore and co-workers¹⁵⁰ utilized genome mining, along with a DNA interference bioassay-guided approach, to investigate two clusters — *ST_pks1* and *spo* — that had been predicted to encode the production of DNA-targeting enediynes. However, genetic inactivation of the *ST_pks1* and *spo* clusters yielded a mutant that was incapable of producing enediyne sporolides but that still exhibited a similar DNA-interfering activity, indicating that enediynes were not responsible for this activity, as earlier predicted¹⁷⁰. Instead, lomaiviticins — glycosylated polyketides produced by another cluster, *ST_pks2* — were found to be solely responsible for the DNA-interfering activity of the parent strain^{150,169}.

A good example of the great potential of genome mining in drug discovery is the biosynthesis of pneumocandin B0 (FIG. 3), a lipohexapeptide from the fungus *Glarea lozoyensis*. Pneumocandin B0 was derivatized to provide its antifungal semisynthetic congener caspofungin, which is currently approved as a treatment against the human pathogenic fungus *Candida albicans*. Decoding the *G. lozoyensis* genome revealed a rich repertoire of new natural-product-encoding genes¹⁵⁵, allowing the engineering of novel pneumocandin derivatives with more-desirable pharmacological properties¹⁵⁵.

The production of anti-infective compounds with novel modes of action has also been explored through metagenomics. Guadinomines (FIG. 3) are produced by *Streptomyces* sp. K01-0509 (REF. 171) and inhibit the type III secretion system (TTSS) of Gram-negative bacteria. Virulence of many pathogenic Gram-negative bacteria — including *E. coli*, *Salmonella* spp., *Yersinia* spp., *Chlamydia* spp., *Vibrio* spp. and *Pseudomonas* spp. — requires the TTSS, which acts as a sensory probe to detect the presence of eukaryotic organisms and to secrete proteins before infection¹⁷². The metagenomics-based

identification and analysis of guadinomine and its biosynthetic pathway has illustrated a means through which a greater yield of guadinomines may be produced¹⁷².

Investigation of metabolites using ‘omics’ approaches. The supply problem of many novel and potent natural products that occur as minor metabolites may be solved by overexpression^{131,161} of their biosynthetic gene clusters in cultivable organisms¹⁷³, although it remains to be seen whether this can be routinely achieved on a production scale.

Overexpression and antisense-suppression strategies have been applied to transgenic plants. For example, the overexpression of several key gene-encoding enzymes — for flavonoids or isoflavonoids¹⁷⁴, or in terpenoid pathways¹⁷⁵ — increases production, thereby circumventing seasonal and geographical limitations. The soybean isoflavone synthase (IFS) was introduced to plants such as tobacco, petunia and lettuce, which do not naturally produce isoflavonoids, enabling these plants to biosynthesize the isoflavone genistein¹⁷⁶. Genistein yield was improved in tobacco by suppressing flavanone 3-hydroxylase (F3H) expression through an antisense approach. F3H is a dihydroflavonol reductase that catalyses the synthesis of dihydroflavonols and anthocyanin. Impairing F3H with the introduction of IFS elevated genistein production. Overexpression of IFS and phenylalanine ammonia-lyase (PAL) in tobacco petals or lettuce leaves also led to an increase in genistein production. Moreover, the presence of both IFS and PAL gave a more substantial increase in genistein yield than in plants in which only IFS was overexpressed¹⁷⁶.

The tobacco plant *Nicotiana benthamiana* has also been explored as a production platform for precursors of artemisinin, which is usually extracted from *Artemisia annua*, for the treatment of malaria¹⁷⁷. Genes for three biosynthetic enzymes necessary to produce artemisinic acid (FIG. 4) were

expressed by agro-infiltration of *N. benthamiana*. This heterologous pathway in the tobacco plant yielded approximately double the artemisinic acid 12- β -D-glucoside than did *A. annua*¹⁷⁸. Keasling and colleagues^{179,180} used the combined expression of *A. annua* artemisinic aldehyde dehydrogenase 1 (ALDH1) and artemisinic alcohol dehydrogenase 1 (ADH1) in engineered strains of *S. cerevisiae* to give a high yield of artemisinic acid. Artemisinic acid is converted to artemisinin in a two-step reaction via reduction of the exocyclic methylene group and simultaneous photo-oxidation¹⁸¹ (FIG. 4). The World Health Organization has recently approved the preparation of semi-synthetic artemisinin that is functionally equivalent to the plant-derived drug¹⁸⁰. Sanofi has taken over the industrial-scale production of semi-synthetic artemisinin to supplement the global supply¹⁸⁰.

Microorganisms such as *E. coli*^{134,140,141} and *S. cerevisiae*^{144,146} have been used as platform hosts for heterologous pathways of plant-extracted flavonoids for nutraceuticals and drug development¹⁷⁴. The bioactive flavonoid 7-*O*-methyl dihydrokaempferol (7-*O*-MeDHK) has been isolated from different plants, but an '*E. coli* cell factory' was established to increase 7-*O*-MeDHK production from its precursor, *p*-coumaric acid¹³⁴. *E. coli* was primarily fed with *p*-coumaric acid to allow synthesis of naringenin (NRN; a related flavanone)¹⁸², which was further derivatized enzymatically to 7-*O*-MeDHK. The flavanone biosynthetic pathway was reconstructed in *E. coli* that expressed three structural genes from three different plant species — 4-coumarate-coenzyme A (CoA) ligase from *Petroselinum crispum*; chalcone synthase from *Petunia hybrid*; and chalcone isomerase from *Medicago sativa* — resulting in a yield of 119 mg NRN per litre of 3 mM *p*-coumaric acid¹⁸³. However, one limitation of the procedure is that *E. coli* produces very low levels of intracellular malonyl-CoA, which is a crucial precursor in the flavanone biosynthetic pathway. The low level of precursor is a potential barrier in using *E. coli* for commercial-scale production of flavonoids, as well as other important polyketides. In a more recent study, the intracellular malonyl-CoA pathway was engineered in *E. coli* by inducing the overexpression of acetyl-CoA carboxylase and acetyl-CoA synthetase genes from *Nocardia farcinica*; this gave a 2.3-fold increase in malonyl-CoA levels in 6 hours and consequently yielded a 2.2-fold increase in the production of NRN over 24 hours upon feeding with 250 μ M *p*-coumaric acid¹³⁴.

In the case of natural products that are sourced from marine invertebrates — such as sponges, soft corals, tunicates and bryozoans — metagenomics technology can promote sustainability and ecological preservation of coral reefs and oceans^{110,114}. Marine organisms have provided many promising bioactive compounds with exciting therapeutic potential, but their development has been severely curtailed owing to the difficulties in obtaining adequate amounts of such compounds. For example, the anticancer agent ecteinascidin-743 (also known as trabectedin; trade name Yondelis; Zeltia/Johnson & Johnson) (FIG. 3) was first isolated from the sea squirt *Ecteinascidia turbinata* in 1984. However, yields from the sea squirt were extremely low: one tonne of animals

was needed to isolate one gram of trabectedin. It was only after 15 years that the supply problem was resolved by a semisynthetic process that is initiated with safracin B, which is obtained by fermentation of the bacterium *Pseudomonas fluorescens*.

Natural products that are derived from marine microorganisms often show pronounced similarities, or are even identical, to compounds from sponges, tunicates or other marine invertebrates. Some of these microorganisms (most of which are currently not cultivable) are now considered to be the true producers of such bioactive constituents¹⁸⁴. The oceans are known to contain an average of 10⁵ to 10⁶ bacteria alone per millilitre of sea water, totaling an estimated bacterial weight of 10¹² tonnes. Genomic mining has played an important part in exploiting both the bacteria and the sponges in which they are found for their biosynthetic genes^{106,110,112,114,185,186}.

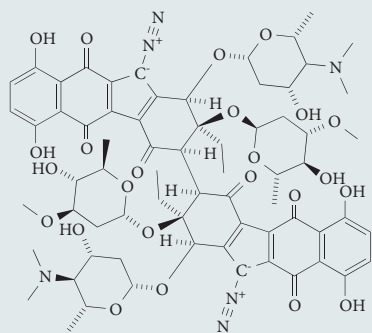
One example of natural products that are synthesised by marine-invertebrate-associated bacteria is the group of antimalarial manzamine alkaloids (FIG. 3) that were originally isolated from the *Acanthostrongylophora* sponges, and later from the sponge-associated actinomycete *Micromonospora*¹⁸⁷. Another example is the group of patellamide peptides (FIG. 3), which are active against multidrug-resistant cancer-cell lines; these peptides were first isolated from the tunicate *Lissoclinum patella*, but were then found to be produced by its cyanobacterial symbiont *Prochloron didemni*¹⁸⁸.

The biosynthetic pathway for the potent antitumour agent psymberin (FIG. 3) from the sponge *Psammocinia* aff. *bulbosa* was obtained through structure-based gene targeting of the biosynthetic polyketide synthase genes from the uncultivated sponge-associated bacteria¹⁸⁴. Strikingly, the analysis of the sponge metagenome suggested that a non-cultivated bacterial symbiont may in fact be the true producer of sponge products, including the marine-sponge-derived polytheonamides^{189,190}. Polytheonamides are ribosomally synthesized peptides and, through metagenomic mining, the biosynthetic scope of ribosomal systems has expanded, opening new doors for peptide and protein bioengineering¹⁹⁰.

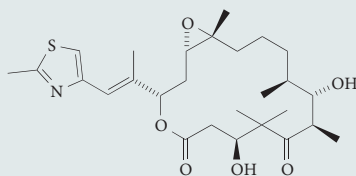
Opportunities for natural products

There are many examples of natural products being used in drug discovery efforts that are directed at a wide range of indications beyond their traditional strengths as antimicrobial and anticancer agents. For example, herbal medicines and isolated compounds have been tested in models of Alzheimer disease^{191–193} and of diabetic neuropathy¹⁹⁴. Here, we focus on two major areas: antimicrobials, and modulators of protein–protein interactions.

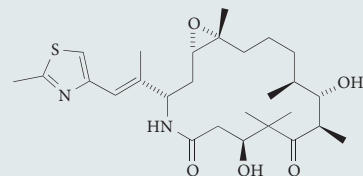
Antimicrobials. Natural products have provided the starting points for most of the major classes of antibiotics, including the β -lactams, aminoglycosides, macrolides, tetracyclines, rifamycins, glycopeptides, streptogramins and lipopeptides. Since 2000, 22 new antibiotics have been launched for treating infections in humans, but only five of these represented new compound classes⁸. Three of these new classes have their origins in natural



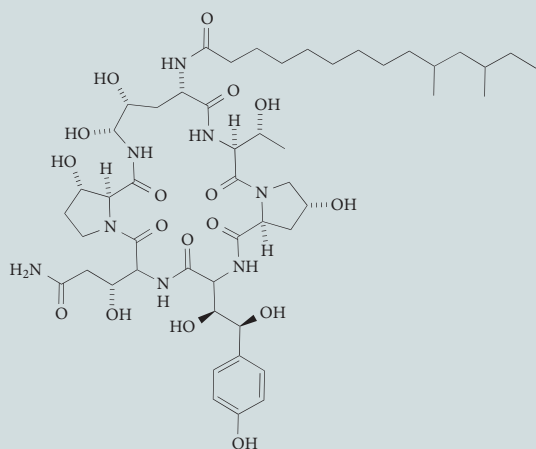
Lomaiviticin A



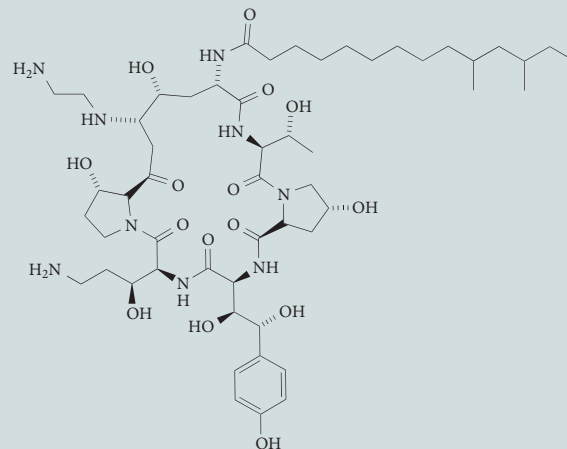
Epothilone B



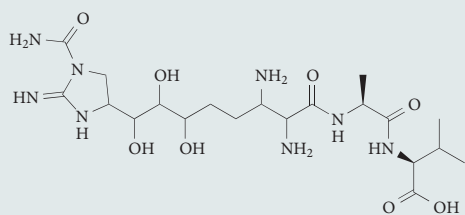
Ixabepilone
(Ixempra; Bristol-Myers Squibb)



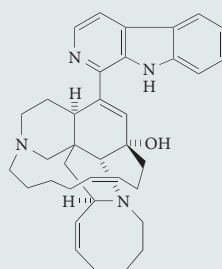
Pneumocandin B0



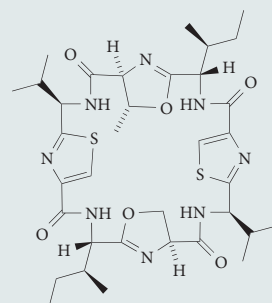
Caspofungin



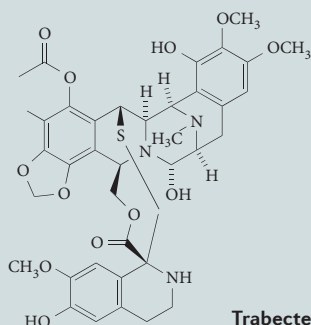
Guadinomine A



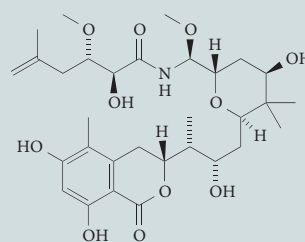
Manzamine A



Patellamide A



Trabectedin
(Yondelis; Zeltia/
Johnson & Johnson)



Psymberin

◀ **Figure 3 | Structures of biologically active natural products or natural-product-derived compounds.** Anticancer lomaiviticins are complex glycosylated diazafluorene polyketides that were originally discovered from *Salinispora pacifica* strain DPJ-I0019 (formerly known as *Micromonospora lomaivitiensis*)¹⁵⁰. The diazobenzo[b]fluorene ring moiety of lomaiviticins interacts directly with DNA²⁸⁷. Ixabepilone (Ixempra; Bristol-Myers Squibb), an analogue of epothilone B, was approved by the US Food and Drug Administration (FDA) in 2007 for the treatment of aggressive metastatic or locally advanced stages of breast cancer. To increase production yield, and to efficiently derivatize a variety of analogues with improved bioactivity, the epothilone biosynthetic gene cluster from *Sorangium cellulosum* was redesigned and reassembled for expression in *Myxococcus xanthus*¹⁶⁸. The pneumocandin biosynthetic gene cluster from *Glarea lozoyensis* (wild-type strain ATCC 20868) was shown to contain a non-ribosomal peptide synthase (GLNRP4); a polyketide synthase (GLPKS4) in tandem duplications; two cytochrome P450 monooxygenases; seven other modifying enzymes; and five contiguous genes for the biosynthesis of L-homotyrosine (a component of the peptide core of pneumocandin B0). Disruption of *GLNRP4* or *GLPKS4* resulted in loss of antifungal activity¹⁵⁵. Guadinomines inhibit the type III secretion system (TTSS), which is responsible for the virulence of many pathogenic Gram-negative bacteria¹⁷¹. The guadinomine gene cluster was identified by targeted disruption of the gene cluster, as well as by heterologous expression and analysis of key enzymes in its biosynthetic pathway²⁸⁸. Manzamine A, patellamide A, trabectedin (Yondelis; Zeltia/Johnson & Johnson) and psymberin are examples of natural products that are produced by marine-invertebrate-associated bacteria. Manzamines are antimalarial alkaloids that were originally isolated from the *Acanthostrongylophora* sponges, and later from the sponge-associated actinomycete *Micromonospora*¹⁸⁷. The patellamide peptides, including patellamide A (shown in the figure), were isolated from the tunicate *Lissoclinum patella* but then found to be produced by its cyanobacterial symbiont *Prochloron didemi*¹⁸⁸. Patellamide peptides are active against multidrug-resistant cancer-cell lines. The anticancer agent ecteinascidin-743 (also known as trabectedin) was first isolated from the sea squirt *Ecteinascidia turbinata* in 1984. Its supply difficulties were later resolved by a semi-synthetic process, commencing with safracin B that was obtained via fermentation of the bacterium *Pseudomonas fluorescens*. The potent antitumour agent psymberin from the sponge *Psammocinia* aff. *bulbosa* has been demonstrated — through the use of structure-based gene targeting — to be produced by the biosynthetic genes of its uncultivated sponge-associated bacteria¹⁸⁴.

products: the lipopeptide daptomycin, the pleuromutilin retapamulin and the tiacumicin fidaxomicin. Butler *et al.*⁸ identified 56 antibiotics that were undergoing clinical trials in 2013. Of these, nineteen represented new structural templates and eleven were related to natural products (TABLE 3). In the past 30 years, natural-product research has also provided the only new class of antifungal drugs — the echinocandins¹⁹⁵.

There is still a pressing need for new and better anti-infectives. The current rate of the introduction of new antimicrobials may not be sufficient to cope with the emergence of bacteria and fungi that are resistant to available agents^{196–199}. The situation is particularly acute in the case of drug-resistant Gram-negative bacterial infections, because very few new compounds are in development for such uses²⁰⁰. The question is whether or not natural products can be useful in finding such agents.

Following the completion of the first DNA sequencing of a bacterial genome in 1995 and the development of genomic technologies, antibiotic discovery switched from traditional functional screening to target-based HTS. Genes that are specific to bacteria were deleted to determine which genes were essential for bacterial viability²⁰¹ and to identify antibacterial targets. However, such efforts have not yielded the success that was anticipated^{201–203}

and no new drugs emerged^{204–206}. A detailed analysis of GlaxoSmithKline's antibacterial campaigns revealed that 67 HTS campaigns against targets that were selected from a panel of 160 genes that had been predicted to be essential for microbial viability failed to generate candidates for clinical development²⁰³. Other companies appear to have had similar experiences²⁰⁴.

The lack of success of the target-based approach is probably due to a combination of three factors. First, identifying functionally essential targets in micro-organisms turned out to be more complicated than had been expected owing to inherent biological complexities. Second, the challenges of moving from hits in a molecular screen to a compound that could reach its intracellular target at an effective concentration were underestimated. Third, the chemical libraries that were used for screening were not sufficiently biologically relevant^{202,204–206}. Moreover, it has been argued that a focus on genomically predicted targets that are single enzymes means that any compounds that are found to be active are liable to trigger drug-resistance mechanisms²⁰⁴.

It is possible that the promise offered by genomic approaches diverted attention from the fact that successful development of antibacterial drugs is always particularly challenging^{202,204,206}. Antibacterials will generally induce the development of resistance mechanisms in the target species, thus rapidly limiting the drug's usefulness. The wide range of drug-resistance mechanisms means that it is increasingly difficult to create new drugs by modifying existing chemical classes. The penetration of compounds through bacterial cell walls and particularly through the additional outer membrane of Gram-negative bacteria is not currently predictable. Compounds also have to avoid the various efflux mechanisms in bacteria. Plasma concentrations of antibiotics generally have to be much higher than those for other drugs in order for the antibiotic to penetrate to its target within the cell, meaning that the antibiotic must have an exceptionally wide safety margin to avoid toxic effects in the patient.

With these obstacles in mind, many research groups are exploring how to make better use of screening that is based on whole organisms, and there is a renewed interest in using natural products in screening. However, there are difficulties associated with screening traditional sources (namely, microbial broths) because of the frequent re-discovery of known compounds. Improvements in rapid dereplication are necessary^{56,58}. Alternatively, other natural products — from different sources — could be tested. These could include microbial products of activated cryptic pathways (see previous section in this Review) or secondary metabolites from plants^{207–212}, endophytic fungi²¹³ or marine sources²¹⁴.

To increase the resolution of screening assays, the sensitivity of target microorganisms for compounds that have particular mechanisms of action can be enhanced by manipulating the levels of a specific target protein or the activity of a certain pathway — for example, through the use of small interfering RNA (siRNA) or genetic engineering. This approach is exemplified by the discovery of inhibitors of the FabF and FabB enzymes (for example,

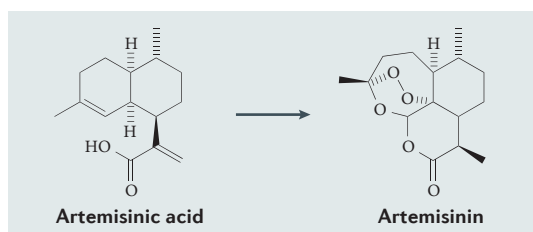


Figure 4 | Structures of artemisinic acid and artemisinin. The antimalarial drug artemisinin (right) is usually extracted from *Artemisia annua*, but yields of artemisinic acid (left) from heterologous expression systems in the tobacco plant are higher. Therefore, it is more economical to convert artemisinic acid to artemisinin via a two-step reaction¹⁸¹.

platensimycin²¹⁵; reviewed in REF. 216) and inhibitors of the cell division protein FtsZ²¹⁷. Collections of bacterial and yeast strains with knockdown or overexpression of particular genes are available, along with information on identified genes that are essential for, for example, *E. coli*²¹⁸, *Staphylococcus aureus*²¹⁹ or *S. cerevisiae*^{220,221}.

Whole-cell screening can be used for determining mechanisms of antimicrobial action by identifying resistant strains, which can be subsequently sequenced to locate functionally important genes²⁰⁵. Using drug-resistant mutants, a group of natural products called acyldepsipeptides were discovered to disrupt a bacterial protease²²²; potentially, these acyldepsipeptides represent a new class of antibiotic. Collections of mutant yeast strains have also been used to uncover mechanisms of action^{220,221}, and such approaches have been successful with natural-product extracts²²³, as exemplified by the screening that led to the isolation and identification of parnafungin from a culture of *Fusarium larvarum*²²⁴. Parnafungin inhibits poly(A) polymerase in *C. albicans* and was found to have broad activity against a range of pathogenic fungi, and also shown to be beneficial in an animal model of candidiasis.

Another functional approach to uncovering novel activities in whole-cell screens involved creating antibiotic mode-of-action profiles (BioMAPs) of 72 known antibiotics against 15 relevant strains of Gram-positive and -negative bacteria³⁶. Subsequently, BioMAP testing of 3,120 pre-fractionated extracts from a marine natural-product collection revealed 83 fractions that produced novel patterns of activity. Novel compounds were isolated, whereas known compounds could be readily identified and disregarded. The BioMAP tool has been made available for use by other researchers³⁶. Information on the activities of 7,500 compounds on yeast cells is also available as a public resource to help find compounds that may be active in phenotypic assays in other model organisms²²⁵.

Simpler approaches to functional screening can still be successful. One example is an assay that identifies inhibitors of bacterial sugar metabolism³⁴, which was used to screen a collection of over 39,000 partially purified microbial extracts from Costa Rica against a sucrose-dependent strain of *Vibrio cholera*. A mutant strain of *V. cholera* that does not transport sucrose was also used

for screening the same compounds, so that bacteriostatic and bactericidal effects could be distinguished. One of 49 initial hits was followed up in detail: it was a novel compound, later identified as 6-propyl gentisyl alcohol. This molecule had activity against other Gram-positive species, but it remains to be seen whether it is non-toxic and whether its rather low potency can be improved by structural modifications.

Other assays have gone beyond tests on the growth of the target microorganisms alone to screening based on infections in model organisms, including in the nematode *Caenorhabditis elegans*²²⁶ and the zebrafish *Danio rerio*^{227,228}. For example, an automated screening system that models *Enterococcus faecalis* infection in *C. elegans*²²⁹ was established to screen natural products in the form of either extracts or pure compounds. The hit rate that was reported for a collection of purified natural products was several times higher than that from synthetic chemical libraries. An advantage of such systems is that the assay can detect activities that treat infections by mechanisms other than classical antibiotic actions.

With the growing awareness of the power of functional assays for antimicrobial activity, and the appreciation of the advantages of natural-product screening libraries, there is likely to be an increase in the use of natural products to find leads with novel antibacterial or antifungal properties. A similar trend has been evident in the interest in seeking compounds that affect interactions between proteins.

Protein–protein interactions. Protein–protein interactions are generally regarded as difficult targets for small molecules^{230–232}, although such interactions have many critical roles in physiology and may therefore represent important therapeutic targets. Screening natural products may be more successful than screening conventional collections of compounds, because the more complex shape and larger size of natural products may make them more likely to perturb interactions between large areas of the involved proteins. As discussed below, several natural products have potent activity at inhibiting or promoting protein–protein interactions. Besides the use of natural products, other approaches to the discovery of drugs that target protein–protein interactions include peptidomimetic design, fragment-based drug discovery and virtual screening, and these have also had some successes (reviewed in REFS 231,233,234).

In theory, inhibitors of protein–protein interactions may have more selectivity than, for example, inhibitors of the active sites of enzymes, because there may be more structural diversity in protein–protein interactions than in active sites. Inhibitors may block interactions by binding to critical regions of one (or more) of the proteins, or via an allosteric mechanism²³². Some early examples of such inhibitors have been described²³⁵. Stabilizers (rather than disruptors) of protein–protein interactions have also been reviewed²³⁶. Some of these stabilizing compounds were found after phenotypic effects that are caused by natural products had been noted and even marketed. Examples of such compounds include paclitaxel (which stabilizes microtubules), rapamycin

Table 3 | Natural products in clinical trials for antibiotic activity that represent new chemical classes*

Compound	Lead structure (chemical class)	Antibiotic activity; possible mechanism of action	Original natural product source	Clinical trial stage
Exeporfinium chloride, also known as XF-73	Porphyrin-based photosensitizer	Gram-positive; membrane disruption	Porphyrin derivative	Phase I study of a nasal gel formulation for reduction of staphylococcal infection in otherwise healthy volunteers (NCT01592214)
NVB302	Deoxyactagardine B (type B lantibiotic)	Gram-positive; inhibition of cell-wall synthesis	<i>Actinoplanes liguriae</i>	Phase I healthy volunteer safety study prior to trials in patients with <i>Clostridium difficile</i> infections [‡]
POL7080	Protegrin I (antimicrobial peptide)	Gram-negative; membrane-pore formation	Porcine leukocytes	Phase II study for ventilator-associated pneumonia linked to <i>Pseudomonas aeruginosa</i> infection (NCT02096328)
LFF571	GE2270A (thiopeptide)	Gram-positive; binds to bacterial-elongation factor Tu, inhibiting translation	<i>Planobispora rosea</i>	Phase II study for moderate <i>Clostridium difficile</i> infections (NCT01232595)
Auriclosene	N-chlorotaurine	Gram-positive, Gram-negative; oxidizing agent	Human leukocytes	Phase II study of an ophthalmic solution for bacterial conjunctivitis (NCT01877694)
GSK1322322	Actinonin (pseudopeptide)	Gram-positive, Gram-negative; inhibits peptide deformylase	<i>Streptomyces</i> spp.	Phase II study for acute bacterial skin infection (NCT01209078)
Brilacidin	N/A (cationic peptide)	Gram-positive, Gram-negative; membrane disruption	Based on defensins, which are found in many vertebrates and invertebrates	Phase II study for serious skin infections (NCT02052388)
LTX-109	N/A (cationic peptide)	Gram-positive, Gram-negative; membrane disruption	Synthetic peptide, based on a pharmacophore identified in lactoferricin B, which was derived from the mammalian iron-chelating protein lactoferrin	Phase II study for Gram-positive skin infections (NCT01223222)
DPK-060	N/A (cationic peptide)	Gram-positive, Gram-negative; membrane disruption	Putative derivative of the antibacterial domain of kininogen	Phase II study of topical application for atopic dermatitis (NCT01522391)
LL-37	37-amino-acid residue from human cathelicidin (cationic peptide)	Gram-positive, Gram-negative; membrane disruption	Human cathelicidin	Phase II trial of a gel formulation for venous leg ulcers [§]

*See REF. 8 for further information. ClinicalTrials.gov identifier included where possible. [‡]See the [Novacta Biosystems clinical trial](#) press release (6 August 2012).

[§]See the [Pergamum clinical trial website](#).

(also known as sirolimus) and tacrolimus (also known as FK506). Both sirolimus and tacrolimus bind to the immunophilin FKBP12. The sirolimus–FKBP12 complex then binds to mammalian target of rapamycin (mTOR), thereby inhibiting the enzymatic activity of mTOR; the tacrolimus–FKBP12 complex binds to and inhibits calcineurin. More detailed information about protein–protein interactions and disruptive compounds is collated in the [2P2I](#) (Protein–Protein Interaction Inhibition) and [TIMBAL](#) databases^{237,238}.

The interaction between the tumour suppressor protein p53 and its regulatory protein MDM2 has served both as a model for protein–protein interactions and as a challenge to find potent and selective inhibitors^{239,240}. p53 regulates the cell cycle in response to stress and its function is dysregulated in many cancer cells, making it an important therapeutic target. Derivatives of the natural product chalcone were revealed by enzyme-linked immunosorbent assays (ELISAs) and NMR assays to disrupt the p53–MDM2 interaction²⁴¹. Screening of a large

collection of microbial extracts led to the identification of chlorofusin as an inhibitor of the p53–MDM2 interaction²⁴². Chlorofusin is a relatively large and complex molecule (it has a molecular mass of 1,363 Da), although it has been synthesized²⁴³. Another compound-collection screen uncovered more potent p53–MDM2 inhibitors, including nutlin 3 (REF. 244). Nutlin 3 has a half-maximal inhibitory concentration (IC₅₀) of 90 nM, and exhibits activities against several cancer-cell lines *in vitro* and in various animal tumour models. Now known as RG7112 or RO5045337, it has completed Phase I trials in various cancers, but has not yet progressed further²⁴⁵. A tryptamine-derived compound (JNJ-26854165, also known as serdemetan) has also been in Phase I trials in patients with solid, refractory tumours²⁴⁶. This compound had previously been shown to be active in several *in vivo* cancer models and has more recently been found to inhibit cholesterol transport in cancer cell lines²⁴⁷, an action that might contribute to its anticancer activity *in vivo*.

Most of the assays that are used to detect the disruption of protein–protein interactions involve sophisticated detection methods such as fluorescence polarization²⁴⁸, surface plasmon resonance and NMR²⁴⁹. A relatively simple and direct assay was described²⁵⁰ that uses phage display with p53 to monitor binding to MDM2. This enabled the rapid screening of a small collection of natural products isolated from mosses and fungi. From this screen, the previously known bioactive compound dehydroaltenusin was identified as an inhibitor of the p53–MDM2 interaction, although it was about 100-fold less potent than nutlin 3 (REF. 250).

Members of the B cell lymphoma 2 (BCL-2) protein family regulate apoptosis. Anti-apoptotic members (such as BCL-X_L), which are commonly overexpressed in cancer cells, dimerize with pro-apoptotic members (such as BAD or BAX). Two natural products, gossypol and purpurogallin, were found to inhibit the binding of BCL-X_L with the α -helical domain of pro-apoptotic proteins²⁵¹. Gossypol and its derivatives inhibit the growth of cancer-cell lines and show anticancer activity in animal models²⁵². Although earlier trials with racemic gossypol were not successful, a single enantiomer of gossypol (called AT-101) is now in clinical trials in patients with different forms of cancer²⁵³. Newer analogues of gossypol are also being tested in cancer²⁵⁴, but it remains to be seen whether or not they will be successful. Thus, the experimental studies with gossypol paved the way to synthetic inhibitors of BCL-2 that are now in clinical trials²⁵⁴.

There are also examples of natural products as inhibitors of protein–protein interactions that have not yet led to clinical development candidates. Thymoquinone and its synthetic derivative poloxin inhibited the interaction between polo-like kinase 1 and its intracellular anchoring site²⁴⁸. These inhibitors induced mitotic arrest and apoptosis in HeLa cells²⁴⁸. Rosmarinic acid and salviolic acids blocked some interactions of the SH2 domain of SRC-family tyrosine kinases with labelled peptides that were designed to mimic the binding domain of the erythropoietin receptor²⁵⁵. An HTS assay that tested for interactions of various proteasome-assembling factors was used to screen a collection of over 123,000 extracts and over 4,000 isolated compounds³⁵. Several potent blockers of the homodimerization of proteasome-assembling chaperone 3 (PAC3) were found. One fungus-derived compound had an IC₅₀ of 20 nM, but its structure was not determined. Another sponge-derived compound, JBIR-22, was later found to be an analogue of equisetin, an antibiotic compound produced by *Fusarium* spp.²⁵⁶. JBIR-22 was shown to be cytotoxic to HeLa cells²⁵⁶, but it was not otherwise characterized for biological activity.

Natural products have also been used to disrupt interactions between proteins and RNA. For example, spliceostatin A (which is a synthetic derivative of a natural product from a broth of a *Pseudomonas* species) blocks splicing and nuclear retention of pre-mRNA, probably by binding to the SF3b complex of the U2 small ribonucleoprotein and blocking its association with the U2 small-nuclear-RNA auxiliary factor²⁵⁷. The spliceostatin A analogue FR901464 kills tumour cells *in vitro*

and prolongs survival in mouse models of cancer²⁵⁸, and simpler spliceostatin A analogues (such as sudemycins) are also being studied as possible anticancer leads²⁵⁹. Didehydro-corticostatin A (a synthetic variant of the steroidal alkaloid corticostatin A from the marine sponge *Corticium simplex*) binds specifically to the transactivation-responsive (TAR) domain of the HIV Tat protein, inhibiting its binding to HIV mRNA²⁶⁰. This prevents transcription, and decreases HIV-1 and HIV-2 replication, and reduces the release of viral particles from CD4⁺ T cells²⁶⁰.

Overall, protein–protein interactions are being recognized as potentially druggable targets, and natural products are likely to provide more leads for future developments.

Conclusions and outlook

Although natural products have been extensively used in historical drug discovery efforts^{3,4}, there are still many resources that could be explored in modern natural-product research^{1,5–7,261}. The Dictionary of Natural Products has recorded approximately 200,000 plant secondary metabolites to date, including about 170,000 unique structures (after the removal of duplicates). Approximately 15% of the drug interventions in the ClinicalTrials.gov database are plant-related, with about 60% of these drugs' sources²⁶² clustered from within only 10 taxonomic families²⁶³. Despite these successes, it is likely that the vast majority of plant species have not been systematically investigated in drug discovery campaigns. Even the traditional plant-based medicines that are used by different cultures still need to be more thoroughly explored.

In addition, microorganisms demonstrate a magnitude of biodiversity that surpasses those of eukaryotes, and can have exceptional metabolic adaptability. For these reasons, microorganisms thrive in even the most extreme environmental conditions. Communities of such microbes exhibit unique prokaryotic diversity and can be considered 'bacterial hotspots'. However, less than 1% of this vast biodiversity has been investigated, mainly owing to non-cultivability in the laboratory. Using metagenomic and heterologous-expression techniques, we can gain better access to a richly diverse microbial community²⁶⁴, and potentially benefit from a boundless source of novel bioactive compounds.

Looking ahead, recent technological advances could be sufficient to revitalize the exploitation of the value of natural products as starting points for drug discovery, particularly with the recent growing interest in phenotypic screening. Understanding the physicochemical properties of natural products in order to facilitate drug development is no different to normal medicinal chemistry principles that are applied to synthetic compounds. The fact that drug-like and lead-like properties can be predicted and experimentally enriched at the fraction-library level³³ allows a front-loading of natural-product drug discovery that is aligned with best-practice medicinal chemistry. Natural products inherently fall in regions of biologically relevant chemical space, as illustrated by recent studies that correlate interactions

in the biosynthetic production of secondary metabolites with similar interactions against validated drug targets. The scaffolds of natural products allow the generation of libraries that escape 'flat land' and retain the highly relevant three-dimensional aspects that are characteristic of natural products. Such three-dimensional attributes of unique natural scaffolds take the generation of chemical libraries into new territory. Whereas the concept of natural-product-inspired scaffolds aims to use chirality and non-flat attributes, natural-product-derived scaffolds have the advantage of using the known protein surface interactions of natural products with biosynthetic enzymes. It remains to be seen how effective the two scaffold strategies will be in the future. In our view, the natural-product–protein interaction that exists in natural-product-derived scaffold mechanisms is more biologically relevant and may prove in the long term to be superior to random three-dimensional (or non-flat) structures in terms of their ability to modulate interactions between proteins that are important in disease states.

This field of research has been enhanced by a rich source of novel compounds from non-traditional sources, such as novel microorganisms. The introduction of recent complex marine natural products to the market²⁶⁵ has also highlighted that process chemistry can deliver sufficient quantities of such molecules if their therapeutic activity is sufficiently compelling. These complex marine natural

products include anticancer drugs such as monomethyl auristatin E, which is a synthetic analogue of dolastatin 10 (REF. 266) from the sea hare *Dolabella auricularia* and a component of the antibody–drug conjugate brentuximab vedotin (Adcetris; Seattle Genetics); eribulin mesylate (Halaven; Eisai), an analogue of halichondrin B²⁶⁷ from the sponge *Halichondria okadae*^{268,269}; trabectedin²⁷⁰ (Yondelis; Zeltia/Johnson & Johnson), a drug derived from the tunicate *Ecteinascidia turbinata*; and the neuro-pathic pain drug ω -conotoxin (also known as ziconotide; (Prialt; Eisai/Jazz Pharmaceuticals))²⁷¹ from the marine snail *Conus magus*. Encouraging more companies to adopt natural-product-based screening, however, requires more natural-product researchers to utilize the recent technological advances effectively.

Note added in proof

A recent paper by Ling *et al.*²⁹⁶ reports on the successful use of an *in situ* cultivation technique to screen extracts from previously uncultured species of soil bacteria for antimicrobial activity. Teixobactin, a new depsipeptide isolated from a species named *Eleftheria terrae*, showed antibiotic activity *in vitro* and *in vivo* against drug-resistant Gram-positive pathogens. Studies indicated a novel mechanism of action involving inhibition of peptidoglycan synthesis, as well as a low propensity for the development of resistance.

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Competing interests statement

The authors declare no competing interests.

FURTHER INFORMATION

Convention on Biological Diversity: <http://www.cbd.int/>
Drug Discovery Portal: <http://www.spiderscience.strath.ac.uk/ddp/>
Nagoya Protocol: <http://www.cbd.int/abs/>
Novacta Biosystems clinical trials: <http://www.novactabio.com/news.php>
Pergamum clinical trials: <http://www.pergamum.com/news/>

DATABASES

2P21 protein-protein interaction inhibition: <http://2p21db.cnrs-mrs.fr/>
AntiBase 2014, The Natural Compound Identifier: <http://eu.wiley.com/WileyCDA/WileyTitle/productCd-3527338411.html>
ChemSpider: <http://www.chemspider.com/>
ClinicalTrials.gov: <http://www.clinicaltrials.gov>
Dictionary of Natural Products: <http://dnp.chemnetbase.com/intro/>
GnPS: Global Natural Products Social Molecular Networking: <http://gnps.ucsd.edu/ProteoSAFe/static/gnps-splash.jsp>
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